

MODULE 3

Machine Learning for Telecom

Prediction, segmentation, anomaly detection and the discipline that makes a model trustworthy enough to act on.

Advanced AI & ML — Telecom Data Analytics

A Nokia - TSSC initiative | Trainer: Ishan Marwah , MSM Technologies



MODULE 3

AIML Foundations

How machines actually learn

AI vs Machine Learning vs Deep Learning

FOUNDATIONS

These three words get used interchangeably. They are not the same thing.

They are nested like Russian dolls. Each one sits inside the one before it.



ARTIFICIAL INTELLIGENCE

Any machine doing something that looks intelligent — including old-fashioned rule engines. **OSS rule: 'If PRB > 80%, trigger alarm'** ✓ AI ✗ No learning

MACHINE LEARNING

The machine works out the rules FROM DATA. Nobody writes them down. Model learns: 'Congestion starts when PRB > 73% AND SINR < 10 dB AND traffic rising'

DEEP LEARNING

ML built on neural networks. A subset of ML multi-layered neural networks for complex pattern recognition

Brilliant at images, audio, language.

Expensive. Hard to explain. Often overkill. ← Day 2

Telecom Mapping: From Concepts to Real Networks



AI Application

Autonomous network optimization systems making decisions across Digital, Network and IT simultaneously without waiting for human input.



ML Application

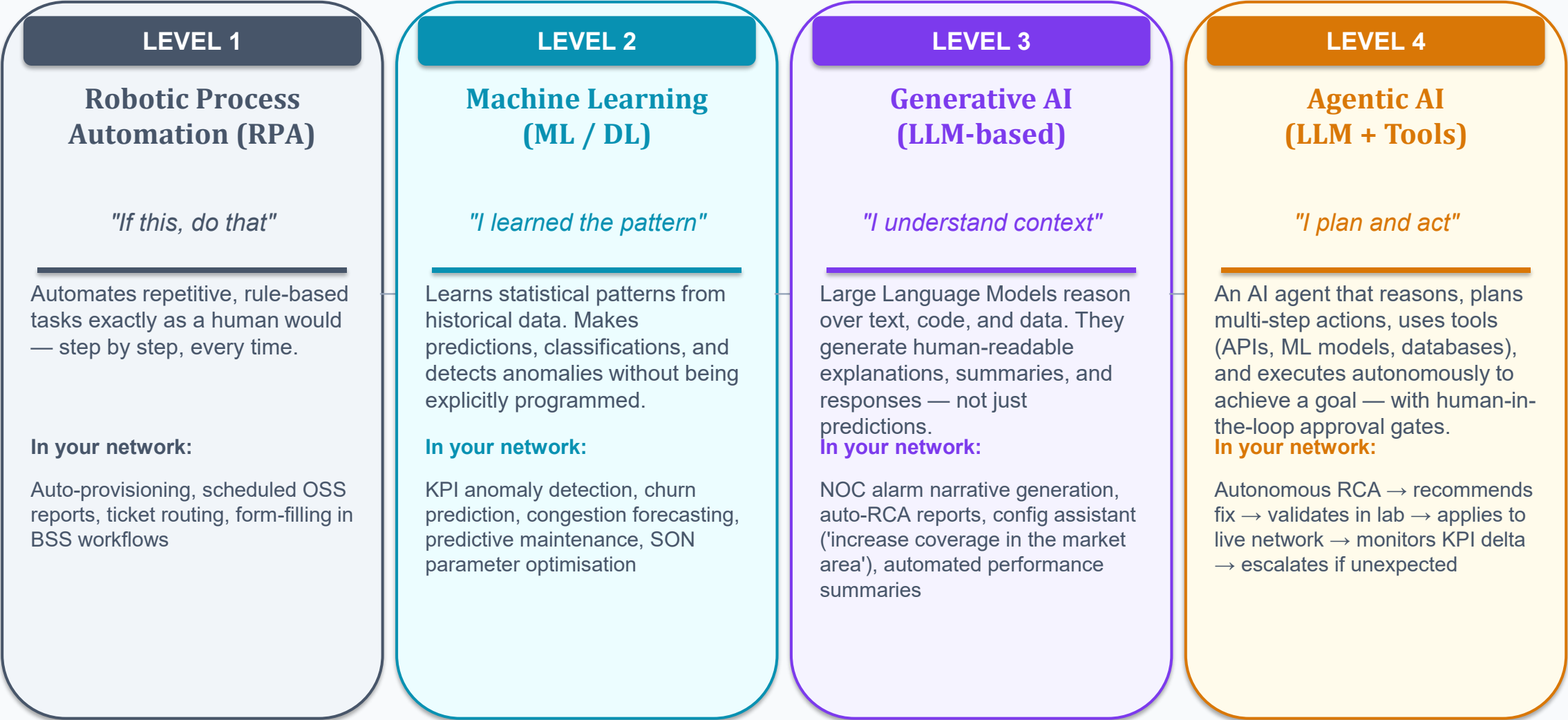
Predicting network congestion from historical KPI trends. The model learns seasonal patterns, busy hours, and growth trajectories — no manual rule-writing.



DL Application

Classifying complex interference patterns directly from raw RF signal measurements using convolutional neural networks what no lookup table could achieve.

The AI Progression Ladder in Telecom



Key insight: These are not competing alternatives — they are complementary layers. A mature telecom AI stack uses all four.

ML Models vs GenAI (LLMs) : How They Differ & Why Telecom Needs Both

Machine Learning Models

answer 'What will happen?'

What it does

Learns statistical patterns from structured telecom data — KPIs, counters, CDRs, alarms

How it learns

Trained on labeled or unlabeled historical datasets. Optimises a mathematical objective (e.g. minimise prediction error)

Output type

A number (regression), a label (classification), a cluster assignment, or an anomaly score

Telecom examples

PRB congestion forecast · Cell failure prediction · Fraud score · SON parameter recommendation · Churn classifier

Strength

Highly accurate on the specific task it was trained for. Fast inference. Works with structured data.

Limitation

Cannot explain itself in plain language. Cannot handle tasks outside training scope. Cannot reason across multiple steps.

VS

Generative AI / LLMs

'What does it mean and what should we do?'

What it does

Understands and generates human language, code, and structured text — using context, reasoning,

How it learns

Pre-trained on massive text corpora using self-supervised learning. Fine-tuned or prompted for specific tasks.

Output type

Natural language narrative, code, structured JSON, a plan, an explanation, or a recommended action sequence

Telecom examples

RCA narrative generation · Config change assistant · Automated NOC reports · Alarm storm summarization · Chatbot for OSS

Strength

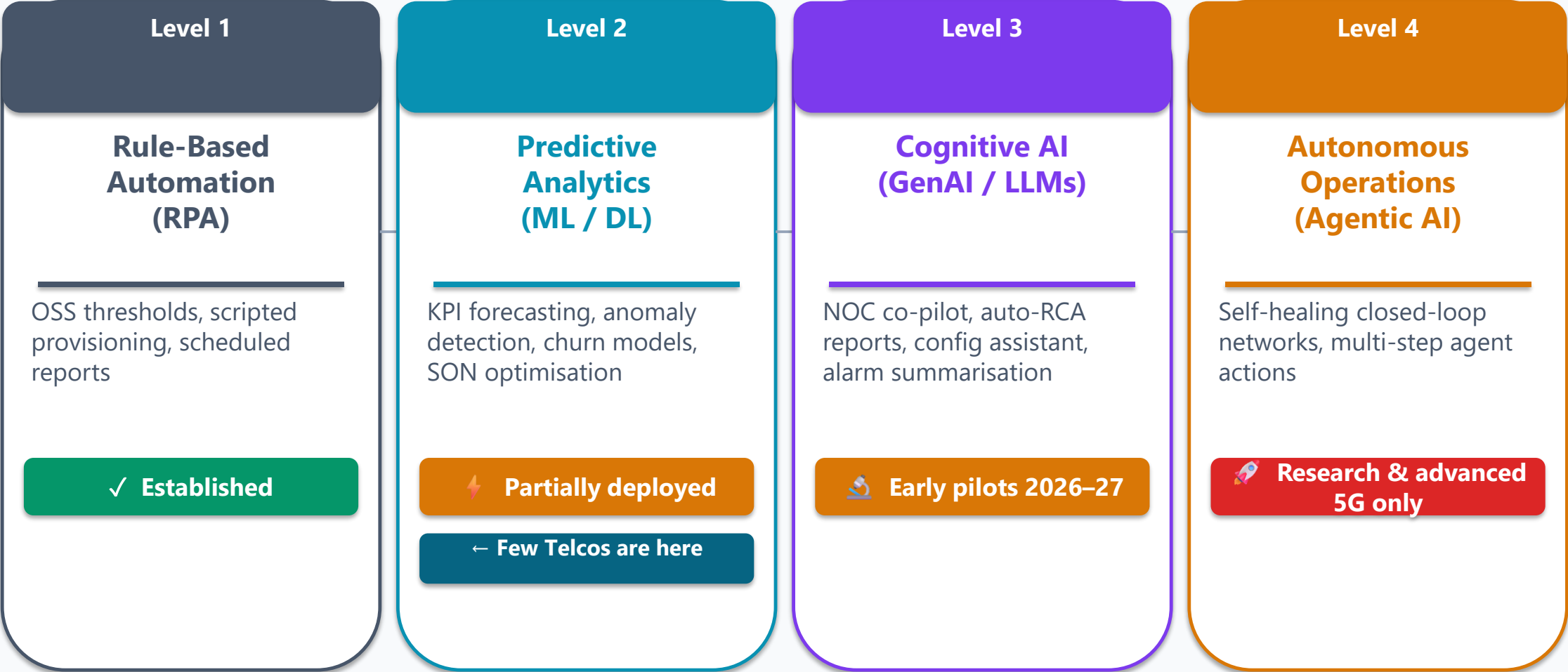
Understands intent and context. Explains ML model outputs in plain English. Bridges technical AI and operational teams.

Limitation

Not designed for precise numerical prediction. Can hallucinate. Needs ML models for quantitative accuracy.

Where Is Your Network on the AI Maturity Curve?

Most operators are between Level 1 and 2 today. The gap to Levels 3 and 4 is closing fast — but only if the foundations are right.



Supervised vs Unsupervised Learning

FOUNDATIONS

The single question that decides everything: do we already know the answers?

8 of you got this right in the pre-assessment.

SUPERVISED · Today

We HAVE the answers from the past.

Learns from labeled data where answers are known.

We know exactly who churned last year.

We show the model the clues AND the answers.

It learns what connects them.

Omantel: churn · revenue forecast · fraud flags

UNSUPERVISED · Tomorrow

We have NO answers. No labels at all.

Finds hidden patterns in data without labels.

Nobody has labelled anything.

We ask the machine: find the structure.

It groups things WE never thought to group.

Omantel: customer segments · odd network behaviour

Quick Check - Supervised or Unsupervised?

POLL

1 We have 3 years of records showing who ported out. Predict who leaves next.

Supervised

2 Group our 2 million subscribers into segments. We don't know what the segments are.

Unsupervised

3 We have 50,000 support tickets already tagged 'billing' or 'network'. Auto-tag new ones.

Supervised

4 Find network cells behaving strangely. Nobody has told us what 'strange' looks like.

Unsupervised

TEAM CHALLENGE

Classify These Network Scenarios

AI (Rule-Based) / ML (Supervised) / ML (Unsupervised) / Deep Learning

1 OSS generates an alarm when cell availability drops below 98%. Threshold set by an engineer.

→ **AI — Rule-Based (no learning)**

2 A system analyzes 3 months of RSRP, CQI, HO success rate to forecast cells with elevated drop rates next week.

→ **ML — Supervised Learning**

3 Deep neural network scans millions of CDR records to identify unusual calling patterns suggesting SIM-box fraud.

→ **Deep Learning**

4 KPI data from 5,000 cells is grouped into natural clusters — nobody told the algorithm how many groups to find.

→ **ML — Unsupervised Learning**

Classification vs Regression

TWO MODEL TYPES

If the output is a number → Regression. If the output is a label → Classification.

	CLASSIFICATION	REGRESSION
The answer is	A CATEGORY	A NUMBER
Example question	Will they churn? Yes / No	How much will they spend? 34.70
Omantel use case	Port-out risk · fraud · ticket type	Revenue · data demand · capacity
We score it with	Accuracy — % we got right	MAE - how many OMR we're off by
Today's exercise	Exercise 1	Exercise 2

Classification or Regression?

POLL

1 Which subscribers will port out to a competitor next month?

CLASSIFICATION

2 How much data will this customer use in Ramadan?

REGRESSION

3 Is this SIM activation fraudulent?

CLASSIFICATION

4 What will our total revenue be next quarter?

REGRESSION



SESSION 3 OF 3

Data & Features

The unglamorous things that decide whether any of this works

Why Data Quality Decides Everything

DATA QUALITY

All models / exercises we have done only worked because the data was clean. In the real world, that is the hard part.



Missing values

A CDR with no duration. A subscriber with no plan. Can't learn from a blank.



Wrong / stale data

A churned customer still marked active. The label lies — the model learns the lie.



Inconsistent units

GB vs MB. OMR vs baisa. Silent — and it wrecks distance-based models.



Imbalance

Fraud is 1.6% of SIMs. A careless model scores 98% by ignoring it entirely.

The columns you feed the model matter more than the model you pick. This is where projects are won.

What it is

Inventing the columns the model actually needs from raw data.

Raw CDRs don't contain **"night_call_pct"** or **"unique_numbers_dialled."**

Someone BUILT those from raw call records. Hence depending on the use case, you built dataset – to get real value from AIML models.

Why it matters most

A great model on bad features loses to a simple model on great features every time.

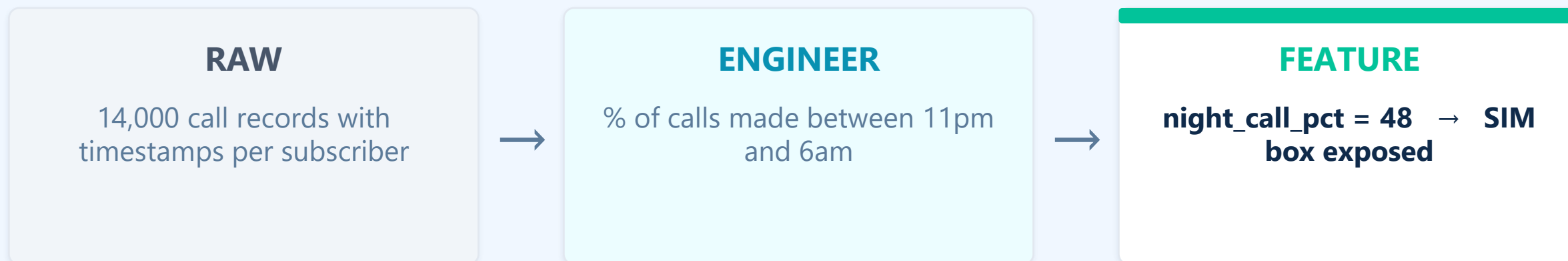
The right feature can make fraud visible that no algorithm could otherwise see.

"night_call_pct" is what exposed the SIM boxes — a raw timestamp never would have.

From Raw Data to a Useful Feature

FEATURES 2/2

The same call records, turned into something the model can actually learn from.



Best practice

Start from a business question, not the data - keep features explainable to your manager/colleagues - one feature = one idea - beware leakage (never use the answer as an input)

Same fraud data. Same model. The **ONLY** change is how we describe the data to it.

✗ RAW DATA

Two SIM activations, as stored:

lat1 23.6 · lon1 58.5
lat2 19.1 · lon2 54.9
minutes_apart 22

The model sees 5 numbers and no way to know they mean 'two places, far apart, minutes apart.'

✓ ONE ENGINEERED FEATURE

We do the maths ourselves:

implied_speed_kmh = 1,240

distance between the two points, divided by the time between them.

No human travels at 1,240 km/h. That SIM is cloned. The feature makes it obvious.

Fraud caught with RAW coordinates: **64%** → with the ONE engineered feature: **100%**

Features in Action — The Midnight Trap

FEATURE POWER 2/2

Sometimes the raw number is not just useless — it actively lies to the model.

✗ RAW: hour as 0–23

Predict network congestion from the hour of day.

The trap:

To the model, hour 23 is a BIG number and hour 0 is a TINY one so it thinks 11pm and midnight are total opposites.

But they're one minute apart. The raw number lies.

✓ ENGINEERED: is_peak_hours

We tell the model what actually matters:

is_peak_hours = 1 if 7pm–1am
= 0 otherwise

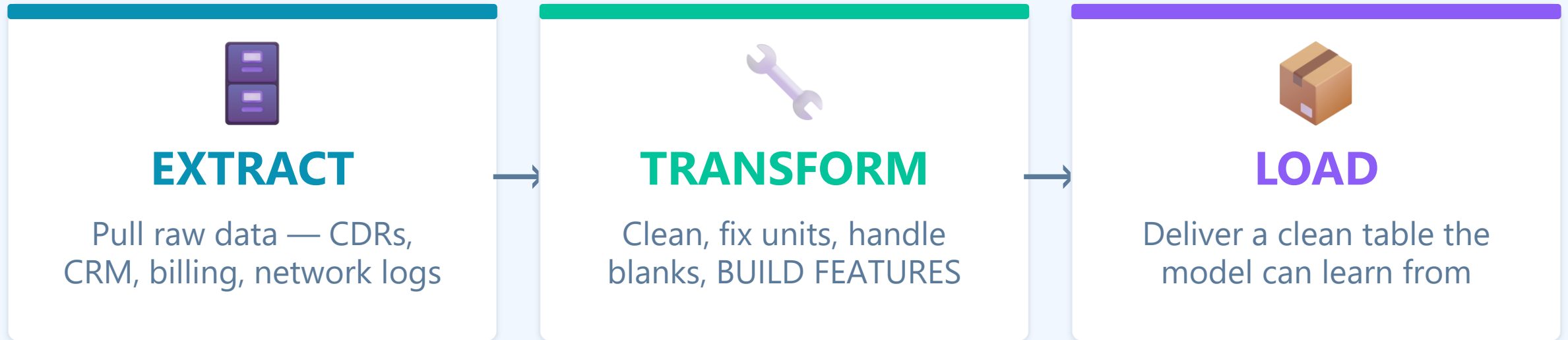
One column that captures the real pattern - the evening + late-night peak instead of a misleading 0-to-23 scale.

Congestion predicted from RAW hour: **68%** → from the engineered flag: **78%**

The ETL Pipeline: How Data Gets Ready

PIPELINE

Before any model runs, raw data has to be pulled, cleaned, and shaped. That plumbing is ETL.



The “T” is where the real work lives. Feature engineering happens inside Transform which is why 80% of a data project is here, not in the model.

Quick Check – These are some of the Data Problem

1

Half the subscribers have no value in the 'plan_type' column.

→ *Missing values*

2

Usage is in GB for postpaid, MB for prepaid, in one table.

→ *Inconsistent units*

3

Customers who left months ago are still marked 'active'.

→ *Wrong / stale data*

4

Turning raw call timestamps into 'night_call_pct'.

→ *Feature engineering (Transform)*



SESSION

Unsupervised Learning

Letting the data group itself

The Two Kinds of Learning — Deeper

FOUNDATIONS

Yesterday you learned the top split. Today each branch splits into two flavours.

SUPERVISED

has an answer key i.e. labels

Classification

predicts a CATEGORY

churn: yes / no → Ex 1

Regression

predicts a NUMBER

ARPU = 34.70 → Ex 2

UNSUPERVISED

no answer key i.e. no labels. Identify hidden patterns

Clustering

groups similar things

K-Means → Ex 3 • TODAY

“But Clustering Gives Labels Too...”

THE TELL

If clustering outputs segment 0,1,2,3 isn't that the same as classification? No and here's why.

CLASSIFICATION

The labels existed first.

Someone said “these people churned.” The model reproduces labels that already existed.

CLUSTERING

The labels did not exist.

The model INVENTED them. Nobody knew there were four groups. That's why YOU name them after.

“Classification puts customers into boxes you already drew.
Clustering draws the boxes.”

Clustering for Network Segmentation



K-Means Clustering

Partitions data into K clusters. Fast and efficient for large datasets.

How it works:

1. Choose K (e.g. K=3 for light/medium/heavy users)
2. Place K random centroids
3. Assign each cell to nearest centroid
4. Move centroids to cluster center
5. Repeat until stable

Telecom use: Customer segmentation, cell performance grouping

Best when: You know roughly how many groups you want



Hierarchical Clustering

Builds a tree of clusters — no need to specify K upfront.

How it works:

1. Each data point starts as its own cluster
2. Progressively merge the closest pairs
3. Continue until all in one cluster
4. Cut the tree at the level that makes sense

Telecom use: Regional network analysis — merge cities with similar patterns into Metro / Tier-2 / Rural groups

Best when: Exploring data without pre-assumptions

Real case: Airtel uses clustering to group cells with similar congestion fingerprints — automatically tailoring SON policy per cluster, not per cell.

Understanding Similarity and Distance

In clustering or segmentation, we don't have labels (like "customer churned = yes/no").

Instead, the model tries to find patterns or groups based on how close data points are to each other.

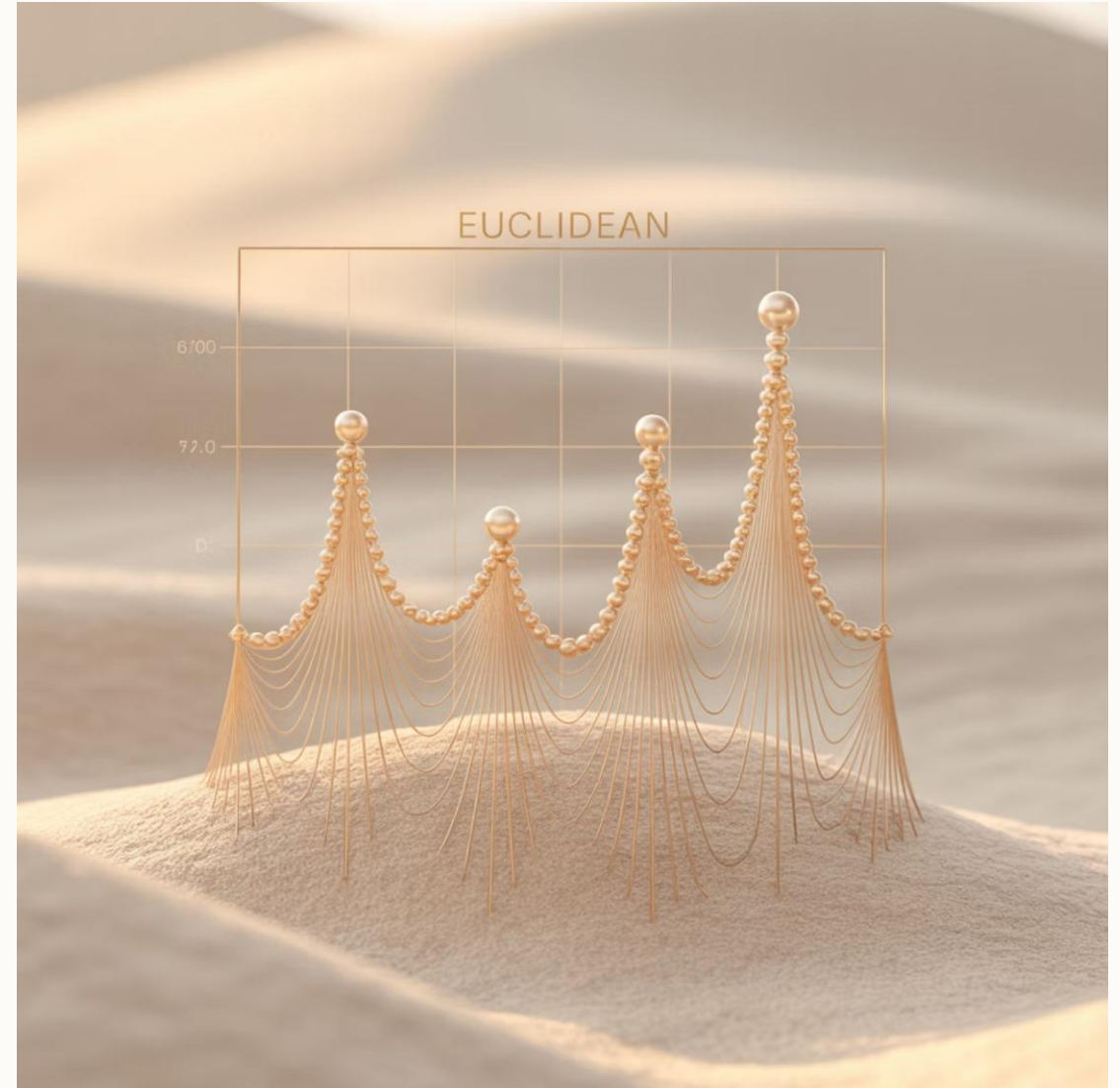
Imagine you have customers represented as points on a chart:

- X-axis = Data Usage (GB)
- Y-axis = Call Minutes

Customers who behave similarly (e.g., both use ~10 GB and make ~400 call minutes) should fall closer together on this graph. Those who behave differently (e.g., 1 GB and 50 minutes) will fall farther apart. So "**similarity**" means "closeness," and the opposite is "distance."

And this is where distance metrics come in

Euclidean Distance is the most common metric—it's literally the straight-line distance between two points in space



The One Thing That Breaks It: Scale

WATCH OUT

K-Means measures distance. If one column's numbers are far bigger, it drowns out all the rest.

Without scaling

voice_minutes swings by 300+. **roaming_days** swings by 5.

Voice shouts, roaming whispers. The model clusters on the loudest column — not the useful one.

With scaling

StandardScaler rescales every column to the same footing.

Now all features speak at the same volume. Its clusters on behaviour, not unit size.

The rule for work: **any model that measures distance needs scaling. Trees don't. K-Means does.**

EXERCISE 3

Segmenting the Subscriber Base

Marketing sends one campaign to 3 million subscribers. Most ignore it — nobody knows who's who.

2,500

subscribers

6

behaviour columns

4

hidden personas

0

labels given

Your job: let K-Means find the groups, then YOU name them and tell Marketing what to do.

Read the Segments Like Four People

One groupby table is the whole exercise. Read each row as a person not as numbers.

Segment	Tenure	Data GB	Spend	Intl	Roam	Who is this?
0	54	22	45	89	2	Loyal High-Value
1	18	62	17	6	1	Heavy-Data Youth
2	14	3	6	3	0	Low-Use Prepaid
3	30	14	28	163	12	Frequent Roamer

Notice: voice_minutes was identical across all four — so the model correctly IGNORED it.

Exercise 3 - What Just Happened

DEBRIEF



Scaling was the whole game

Unscaled - nonsense. Scaled - real customer types.



You named them, not the model

K-Means gave 0,1,2,3. You made them meaningful.



It rediscovered hidden truth

Recovered 4 personas it was never shown from behaviour.



The output is a business action

One campaign per segment. No machine can do that step.

“A new customer activates a SIM tomorrow. Re-run K-Means on 3M rows to find their segment?”

1

DISCOVER

Run K-Means once. Find 4 segments exist. Name them.

Unsupervised - today

Once a quarter

2

OPERATIONALISE

Train a classifier on those labels. Every new SIM assigned instantly.

Supervised - Day 1

Every new subscriber

Deep Learning - What It Actually Is

"Deep" doesn't mean "smart" or "advanced" in some vague marketing sense.

The plain-English version

A neural network is layers of simple switches passing signals forward.

Each layer learns to spot a slightly bigger pattern than the one before it. Stack enough layers and it can recognise very complex things — a face, a spoken word, the meaning of a sentence.

Nobody programmed any of these definitions of layer. They learned from data, to pick each layer's role

Why depth helps

Layer 1 finds edges

Layer 2 combines edges into shapes

Layer 3 combines shapes into a face

Each layer builds on the last. That stacking is the "deep" in deep learning.

Deep Learning — How It Actually Works

First — a neural network is NOT the same as deep learning. **Deep learning = a neural network with MANY stacked layers.**

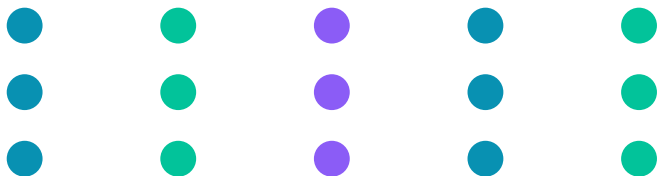
Neural network vs deep learning

SHALLOW (1–2 layers)



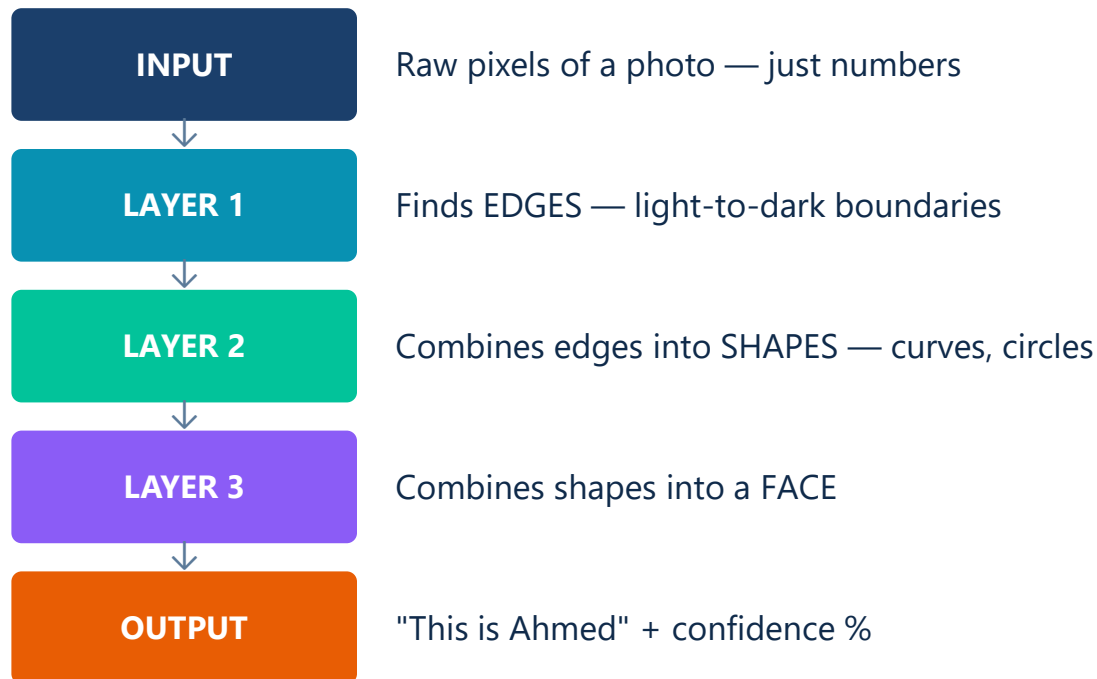
Still a neural network — just not 'deep'.

DEEP (many layers)



More layers = 'deeper'. That stacking IS the word 'deep'.

How the layers build on each other



Nobody wrote these rules. Each layer LEARNED its job from thousands of examples.

Deep Learning - What It Actually Is

"Deep" doesn't mean "smart" or "advanced" in some vague marketing sense.

Stacking

A "shallow" network might have one or two layers. A "deep" one might have dozens, each one working on a progressively more abstract version of the pattern the layer before it found.

more layers means more computation, more training data needed, and more risk of the kind of overfitting problem you already covered on Day 1.

LLMs are built from this exact same idea of stacked layers wildly different scale.

Why depth helps

"So more layers is always better?"

Depth is a tool matched to problem complexity, not a scoreboard.

"is this what powers ChatGPT ?

Each layer builds on the last. That stacking is the "deep" in deep learning.

How Many Layers? Who Decides?

There is no magic number — and no official threshold. **You add layers until the problem is solved, then stop.**

How many is 'deep'?

No official rule. Loosely, more than 2–3 hidden layers gets called 'deep'. Nobody polices it. A face-recogniser may use dozens; a language model over a hundred.

Who decides?

The engineer — based on how COMPLEX the pattern is. Simple pattern → few layers. Recognising faces or understanding language → many. It's matched to the problem, not chosen for show.

Is there a limit?

Yes — three ceilings: COST (each layer adds compute + time), DATA (more layers need more examples or they overfit), and DIMINISHING RETURNS (past a point, extra layers stop helping).

The rule that matters

More layers is NOT better — ENOUGH layers is better. For the spreadsheet-shaped data in your churn and revenue exercises, ZERO deep-learning layers was the right answer. That's not a limitation — that's choosing the right tool.

Deep Learning vs What You Built

Two honest differences — and why simpler was the RIGHT choice for today's data.

Today's models		Deep learning
Works best on	Tables (rows & columns) – Structured Dataset	Images, audio, language – Unstructured Dataset
Features	You engineer them	It learns them itself
Explainable?	Yes - you can show why	Usually a black box
Cost to run	Cheap, fast	Expensive, slow, data-hungry

For a spreadsheet of subscribers, simpler models are faster, explainable, and just as accurate.

Where Deep Learning Earns Its Keep

Three telecom places it's genuinely worth it — and the one that opens Day 3.



RF & images

Reading interference patterns from raw signal maps no lookup table could match.



Speech

Transcribing and understanding call-centre audio at scale.



Language

Reading customer feedback and support tickets — turning text into meaning.

Tomorrow → **we make a machine READ.** Text becomes tokens → tokens become numbers (embeddings) → numbers carry meaning.

That path — tokens → embeddings → meaning — is deep learning applied to language. It's the whole of Day 3, Session 1.

Quick Check — Deep Learning or Not?

POLL · DEEP LEARNING

Hands up: is deep learning the RIGHT tool — or is a simpler model better?

1 Predict churn from a table of subscriber KPIs. → *Simpler model — it's a table*

2 Read raw RF images to spot interference. → *Deep learning*

3 Understand what a customer's complaint text means. → *Deep learning (language)*

4 Forecast next month's ARPU from billing history. → *Simpler model — it's a table*

NEXT

Module 4 — Forecasting & Real-Time Analytics

You can now predict whether something will happen. Module 5 asks when, and how confident you are — KPI forecasting with uncertainty, seasonality in telecom traffic, and monitoring a live KPI on a dashboard rather than in a notebook.

Before the next session: complete MoP 4.2, MoP 4.4 and Mini-Capstone 2. Bring your tuned churn model — Module 5 forecasts the KPI it predicts against.